Aydin Javadov

PhD Candidate at ETH Zurich

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Education

PhD Candidate in Machine Learning *Nov. 2024 – Present*
ETH Zurich Zurich, Switzerland

Associated Researcher at ETH AI Center. Focusing on: Human-AI Alignment, Explainable AI, Multimodality, Large Language Models.
Supervised by: Prof. Dr. Florian von Wangenheim (Mobiliar Lab for Analytics @ ETH Zurich) & Prof. Dr. Bjoern Schuller (GLAM @ Imperial College London & CHI @ TUM).

M.Sc. in Data Engineering & Analytics; Graduation with Distinction (Top $\leq$ 10%) *Apr. 2021 – Jul. 2024*
Technical University of Munich Munich, Germany

German Grade: 1.5.
Thesis (conducted during research stay at BMW Research):
Explainable AI for Graph Representation Learning and Clustering Algorithms (Grade: 1.0).
Supervised by: Prof. Dr. Viktor Leis.

Exchange Stay — Computer Science *Feb. 2018 – Jun. 2018*
Korean Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea

B.Sc. in Computer Engineering; Graduation with Distinction (Top $\leq$ 1%) *Sep. 2016 – Jun. 2020*
ADA University Baku, Azerbaijan

US GPA: 3.89 (German Equivalent: 1.1).
Thesis: *Analysis of Distributed Temperature Sensor Data Using Machine Learning.*
Supervised by: Prof. Dr. Samir Rustamov.

Experience

PhD Candidate in Machine Learning & ETH AI Center Associated Researcher *Nov. 2024 – Present*
ETH Zurich Zurich, Switzerland

- Supervised by Prof. Dr. Florian von Wangenheim (Mobiliar Lab for Analytics @ ETH Zurich) and Prof. Dr. Bjoern Schuller (GLAM @ Imperial College London & CHI @ TUM).
- Focusing on: Human-AI Alignment, Mechanistic Interpretability, Multimodality, Large Language Models.
- Reviewer at Apertus Claritas, interpretability platform for Apertus, Switzerland's open multilingual LLM.

Machine Learning Research Student *Apr. 2023 – May 2024*
BMW Group Munich, Germany

- Led the project on Explainable AI integration into graph learning for engine-problems; pushed to the production pipeline.
- Focusing on: Large Language Models, Time Series Analysis, Graph Representation Learning, Explainable AI.

Master's Thesis Research *Oct. 2022 – Apr. 2023*
BMW Group Munich, Germany

- Thesis: *Explainable AI for Graph Representation Learning and Clustering Algorithms.*
- Technologies: Python, PyTorch, AWS, Git. Grade: 1.0 (German System).

Guided Research — Chair of Computational Imaging and Inverse Problems *Apr. 2022 – Nov. 2022*
Technical University of Munich Munich, Germany

- Project: *Explainable AI and Computer Vision for Clinical Decision Support in Dermatology.*
- Understanding and implementation of interpretability techniques for deep learning models for skin lesion classification in close collaboration with Munich University Clinic physicians.

Data Science Working Student *Sep. 2021 – May 2022*
novuter GmbH Munich, Germany

- Extracted raw data and transformed data stories across domains (e.g., finance) to optimise decision-making.
- Technologies: PostgreSQL, JavaScript, Python, R.

Artificial Intelligence Intern *Oct. 2019 – Feb. 2020*
ATL Tech – AI Lab Baku, Azerbaijan

- Advanced research on Speech Recognition in Dialog Systems for the Azerbaijani language.
- Technologies: Python, pandas, NumPy.

Selected Publications

- **EMNLP 2026:** R. Wu, F. Draye, **A. Javadov**, B. Schölkopf, T. J. Zhang, Z. Jin. *Computation Graph Recovery from Chain-of-Thought.*
- **ICLR 2026:** Y. Wang, Y. Hou, **A. Javadov**, M. Akhtar, M. Sachan. *Compose and Fuse: Revisiting the Foundational Bottlenecks in Multimodal Reasoning.* [arXiv]
- **COLM 2026** (Actionable Interpretability Workshop) **A. Javadov** *When Are Routing Weights Actionable? Causal Probes for Interpreting Block Attention Residuals* [arXiv]
- **COLM 2026** (Agentic Behavior Workshop) **A. Javadov**, S. Aitkazinov, T. Hoesli, F. Wangenheim, B. Schuller, J. Ollier. *NeuReasoner: Theory-grounded Mapping of Reasoning Elicitation Boundaries* [arXiv]
- **Frontiers in AI, 2026:** Q. Sun, Y. Li, **A. Javadov**, X. Wu, B. W. Schuller. *Prior-Aligned Frequency-Domain Explanations for Heart Sound Classification: A Scale-Consistent Attribution Approach.* [Paper]
- **NeurIPS 2025** (Time Series for Health Workshop): **A. Javadov**, S. Garibov, T. Hoesli, Q. Sun, F. von Wangenheim, J. Ollier, B. Schuller. *RAxSS: Retrieval-Augmented Sparse Sampling for Explainable Variable-Length Medical Time Series Classification.* [arXiv]
- **CHI 2025 Workshop** (Envisioning the Future of Interactive Health): **A. Javadov***, M. Baumgartner*, R. Ney, F. von Wangenheim, J. Ollier. *BioSyncHRI: Synchronizing Human Robot Interaction via Real-Time Biosignal Adaptation.*
- **CHI 2025 Workshop** (4th Generative AI and HCI): *Generative AI for Wellness Applications via User-Generated Immersive Virtual Environments.*
- **IEEE IDS'25** (Special Track: FinRLFM): E. Alturki, **A. Javadov**, B. Schuller. *Adaptive Confidence-Weighted LLM Infusion for Financial Reinforcement Learning.* [Paper]
- **Applied Intelligence, 2021:** J. Hasanov, S. Garibov*, **A. Javadov***. *Approximation of CIEDE2000 Color Closeness Function Using Neuro-Fuzzy Networks.* [Journal]

Awards & Participations

- **84th out of 78,545 students (top 0.001%):** National University Entrance Exam of Azerbaijan – July 2016.
- **Winner:** Hackathon HackaTUM, TUM & Carl Zeiss AG (ZEISS) – November 2021.
- **228th out of 40,875 students (top 0.005%):** Primary School Republic Examination of Türkiye – November 2010.
- **Global Korea Scholarship:** Ministry of Education of the Republic of Korea – Feb.–Jun. 2018.
- **2× Rector's List of Honour & 3× Dean's List of Honour:** Merit-Based Scholarship at ADA University – 2017–2020.
- **Organiser of 'Purple Comet' International Math Olympiad:** ADA University – April 2019.

Technical Skills

- **ML/AI Concepts:** Representation Learning, LLMs, Explainable AI, Deep Learning, Computer Vision, Time Series, Deep Generative Models, Uncertainty Quantification.
- **Application Areas:** Medicine, Cognitive Modelling, Human Factors, Finance, Signal Processing, Dynamic Systems.

- **Tools:** Python, PyTorch, Git, Bash, AWS, Azure, Spark.

Languages

- **Native:** Turkish, Russian, Azerbaijani.
- **Fluent:** English.
- **Elementary:** German.
- **Beginner:** Korean.

Referees

- **Academic:** Prof. Dr. Bjoern Schuller (Imperial College London & TUM); Prof. Dr. Viktor Leis (TUM); Prof. Dr. Felix Dietrich (TUM); Prof. Dr. Florian von Wangenheim (ETH Zurich).
- **Industry:** Prof. Dr. Tobias Lasser (NVIDIA); Dr. Raphael Weingartner (BMW); Dr. Iryna Bastian (BMW).